

Marginalized Stacked Denoising Auto-Encoders for Mixed-Type Data Clustering

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Abstract

Clustering mixed-type data which comprises both numerical and categorical features remains a critical yet underexplored challenge in unsupervised learning. In this paper, we propose a novel framework that integrates multi-view similarity fusion with a graph-regularized marginalized stacked denoising autoencoder (mSDA) to address the limitations of existing methods. Our approach constructs feature-type-specific similarity matrices using RBF kernels for numerical attributes and cosine similarity for categorical variables. These matrices are fused into a unified affinity structure that guides the mSDA encoding via closed-form layer-wise denoising, followed by non-linear activation to enhance expressivity. Unlike prior deep models, our method does not require an explicit graph or iterative optimization, making it both scalable and versatile. We evaluate our model on a suite of real-world and synthetic mixed-type datasets and compare it against state-of-the-art baselines such as k-SubMix. Quantitative results using metrics like NMI, ARI, and Silhouette Score demonstrate consistent performance improvements, particularly in heterogeneous and imbalanced scenarios. Furthermore, qualitative visualizations (e.g., similarity heatmaps and UMAP projections) highlight the ability of our learned representations to uncover meaningful latent structures. Our findings underline the importance of similarity-aware, type-sensitive representation learning for robust clustering in tabular data.

Introduction

Context

Clustering is a fundamental task in unsupervised machine learning, aiming to uncover latent groupings within unlabeled datasets (Jain, Murty, and Flynn 1999). Classical algorithms such as K-Means (Lloyd 1982), Spectral Clustering (Von Luxburg 2007), and DBSCAN (Ester et al. 1996) perform well on homogeneous data but often falter when faced with mixed-type datasets which contain both numerical and categorical features. This scenario is common in applications ranging from healthcare to finance and social sciences (Ahmad and Khan 2019).

A major challenge lies in defining similarity measures that faithfully capture relationships between continuous and

discrete attributes. Simple strategies, like one-hot encoding or naive concatenation, distort data geometry and degrade clustering performance (Hastie, Tibshirani, and Friedman 2009). While specialized methods such as k-Prototypes (Huang 1997), Gower’s distance (Gower 1971), and subspace models like K-Submix (Elhamifar and Wang 2020) have been developed to address mixed data, they often require careful tuning and may not generalize well to complex, real-world datasets.

Motivation

The prevalence of heterogeneous data in practical applications underscores the need for clustering methods that natively handle mixed feature types. While deep learning approaches like the marginalized stacked denoising autoencoder (mSDA) (Chen et al. 2012) have advanced representation learning, their application to mixed-type tabular data remains limited. Most frameworks lack mechanisms to preserve cross-feature-type relationships during representation learning.

Hence, this work bridges three critical gaps:

- **Adaptation of mSDA** for mixed data through feature-type-specific processing.
- **Multi-view similarity fusion** combining numerical (RBF) and categorical (cosine) metrics, as an alternative to classical measures like Gower’s distance.
- **Graph-regularized denoising** using sample similarity to preserve local structures.

Unlike graph autoencoders requiring explicit adjacency, our approach derives similarity directly from feature interactions, making it suitable for tabular data. We rigorously evaluate our method against state-of-the-art baselines using established mixed-data benchmarks.

Problem

Current denoising autoencoder frameworks, including mSDA (Chen et al. 2012), operate under the assumption of homogeneous input data. This limits their effectiveness on mixed-type datasets where features exhibit fundamentally different statistical properties. Key challenges include:

- **Cross-Modal Integration:** No mechanism to preserve relationships between numerical and categorical features during denoising.

- **Similarity Metric Fragmentation:** Separate treatment of feature types obscures cross-type dependencies.
- **Representation Distortion:** Concatenating features before encoding fails to capture type-specific semantics.

Our solution addresses these limitations through:

1. Multi-view similarity fusion with parametric weighting ($S_{fused} = \alpha \cdot S_{num} + (1 - \alpha) \cdot S_{cat}$)
2. Graph-informed denoising, where S_{fused} guides the covariance computation in each mSDA layer
3. Joint optimization of multi-level feature-type-specific representations via stacked encoding

Contribution

In this work, we propose a novel extension of the marginalized stacked denoising autoencoder (mSDA) framework, specifically designed for mixed-type tabular data. Our approach integrates both numerical and categorical features through a principled multi-view similarity fusion, enabling the model to capture complex relationships inherent in heterogeneous datasets. By leveraging graph-based regularization, our method uses fused similarity matrices to guide the denoising process and preserve local sample relationships, all without requiring an explicit graph structure as input. To rigorously assess the effectiveness of our framework, we conduct comprehensive experiments on established mixed-data benchmarks, comparing our results to state-of-the-art methods such as K-Submix. Our evaluation spans multiple datasets and employs both supervised and unsupervised clustering metrics, with systematic hyperparameter optimization via grid search. This comprehensive analysis demonstrates the robustness and versatility of our approach for clustering in real-world mixed-type data scenarios.

Related Work

Clustering mixed-type data poses unique challenges that have been the focus of extensive research across both traditional and deep learning paradigms. Early methods such as k-Prototypes (Huang 1997) and Gower’s distance (Gower 1971) were specifically designed to handle heterogeneous attributes by combining dissimilarity measures. While these approaches remain influential, they often rely on heuristic definitions of similarity and may not scale well or generalize to complex, high-dimensional datasets (Ahmad and Khan 2019).

More recently, subspace-based techniques like K-Submix (Elhamifar and Wang 2020) have shown promise by leveraging locally linear subspaces for each cluster. Though effective, these methods require careful regularization and hyperparameter tuning, and they lack the capacity for representation learning, which is a key advantage in modern unsupervised pipelines.

Denoising autoencoders (DAEs) (Vincent et al. 2008) have provided a deep learning alternative by learning robust latent representations. Marginalized Denoising Autoencoders (mDA) (Chen et al. 2012) extended this idea with a closed-form solution, enabling efficient layer stacking (mSDA) and bypassing iterative training. However, mSDA

was originally developed for homogeneous inputs and does not account for type-specific correlations between features.

Building on this foundation, (Fettal, Labiod, and Nadif 2023) proposed an extension of mDA for domain adaptation with partially observed data. Their framework jointly marginalizes both the feature corruption and the missingness pattern, enhancing robustness in transfer learning settings. While their objective differs (supervised domain adaptation), the underlying motivation to generalize mDA to more realistic input structures aligns with our goal of adapting mSDA to heterogeneous tabular data. This demonstrates the broader versatility of the mDA framework across modalities.

To incorporate graph-based structure into representation learning, the Marginalized Graph Autoencoder (MGAE) (Wang, Cui, and Zhu 2017) introduces Laplacian regularization within an mDA framework. Although effective on structured data (e.g., citation networks), it assumes an existing adjacency graph and uniform feature types; assumptions that do not hold in general tabular settings.

Our work builds on these foundations by adapting mSDA to explicitly handle mixed-type tabular data. Inspired by multi-view learning principles, we propose a similarity fusion strategy that integrates RBF kernels for numerical variables and cosine similarity for categorical features. Unlike traditional approaches that concatenate features or require handcrafted similarity functions, our method learns fused similarities directly from data and integrates them into a graph-regularized encoding framework.

Surveys such as (Ahmad and Khan 2019) highlight the scarcity of deep clustering models tailored for mixed data. By incorporating feature-type-specific processing and scalable multi-layer representations, our model addresses these gaps and offers a principled alternative to existing baselines.

Methodology

Pipeline Overview

Our proposed pipeline operates in five main stages: (1) pre-processing and encoding of mixed-type tabular data, (2) construction of numerical and categorical similarity matrices, (3) multi-layer feature transformation using marginalized stacked denoising autoencoders (mSDA) with non-linear activations, (4) clustering with Spectral Clustering, and (5) evaluation with internal and external clustering metrics.

Multi-View Similarity and Graph-Regularized Denoising via mSDA

To capture both intra-modality structures and inter-sample relationships, we construct a fused similarity matrix S_{fused} based on feature-type-specific similarity measures. Numerical features (standardized using `RobustScaler` for real-world datasets and `StandardScaler` for synthetic ones, depending on the presence of outliers) are compared via a Gaussian RBF kernel, while categorical variables (one-hot encoded) are compared using cosine similarity:

$$S_{num} = \text{RBF}(X_{num}, \gamma), \quad S_{cat} = \text{cosine}(X_{cat})$$

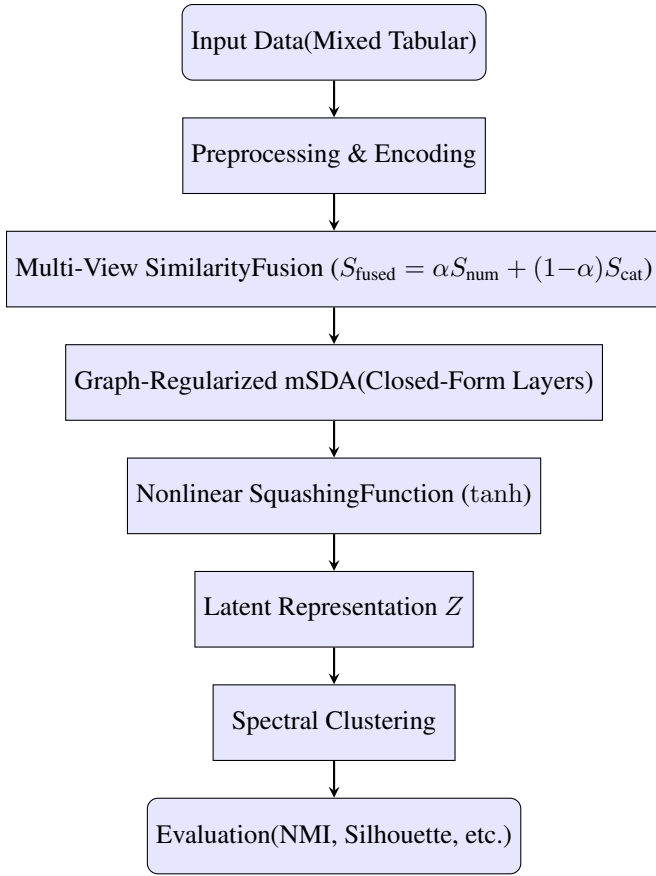


Figure 1: Pipeline overview: from mixed-type input data to evaluation via nonlinearly-augmented mSDA and spectral clustering.

Each similarity matrix is row-normalized to ensure stochasticity. The final fused similarity is computed as a convex combination:

$$S_{\text{fused}} = \alpha S_{\text{num}} + (1 - \alpha) S_{\text{cat}}$$

where $\alpha \in [0, 1]$ controls the relative contribution of each view.

This fused matrix acts as a graph surrogate to regularize the marginalized stacked denoising autoencoder (mSDA). Instead of an explicit adjacency matrix, S_{fused} encodes sample-level similarities and guides the denoising process.

Marginalized Corruption. Unlike standard Denoising Autoencoders (DAEs), mSDA does not explicitly generate multiple corrupted versions of the input data. Instead, the corruption process is analytically marginalized: the reconstruction weights W are computed to minimize the expected squared reconstruction loss under random feature corruption:

$$E_{\tilde{X} \sim q(\tilde{X}|X)} [\|X - W\tilde{X}\|^2]$$

As shown by Chen et al. (Chen et al. 2012), this expecta-

tion yields a closed-form solution based on two scatter matrices:

$$P = E[\tilde{X}X^\top], \quad Q = E[\tilde{X}\tilde{X}^\top], \quad \text{so that} \quad W = P(Q + \lambda I)^\dagger$$

The entries of P and Q are adjusted by the probability $1 - p$ of a feature surviving the corruption. In our implementation, this expectation is encoded directly into the construction of P and Q using a corruption mask q , avoiding the need for sampling. This marginalization enables fast and deterministic training without stochastic optimization or sampling noise.

mSDA Architecture with Nonlinearity. At each layer ℓ , mSDA computes a closed-form solution:

$$W^{(\ell)} = P(Q + \lambda I)^\dagger$$

where P and Q are scatter matrices weighted by S_{fused} , and λ is a regularization coefficient. A corruption mask q controls feature noise injection, and a bias term is explicitly added:

$$X_{\text{bias}} = \begin{bmatrix} X \\ \mathbf{1} \end{bmatrix}$$

The intermediate latent representation before nonlinearity is:

$$\tilde{Z}^{(\ell)} = W^{(\ell)} X_{\text{bias}} S_{\text{fused}}$$

A non-linear activation function is then applied element-wise:

$$Z^{(\ell)} = \tanh(\tilde{Z}^{(\ell)})$$

This step increases the expressivity of the learned features while preserving the closed-form training of mSDA. We empirically found that adding non-linearity after each layer improves clustering quality.

Final Representation. The output representation is obtained by concatenating all nonlinear encodings:

$$Z = [Z^{(1)} \mid Z^{(2)} \mid \dots \mid Z^{(L)}]$$

This graph-regularized, non-linear stacked encoding captures both local neighborhood structure and global feature interactions, making it well-suited for clustering mixed-type data.

Clustering and Evaluation

The final representation Z is used as input to a **Spectral Clustering** algorithm. The number of clusters is fixed to the number of ground-truth classes. Depending on the dataset characteristics, we select the affinity function as follows:

- **RBF affinity** is used when the data exhibits globally smooth variations (e.g., *Heart*, *Adult* datasets).
- **Nearest neighbors affinity** is used for datasets with locally clustered or irregular structures (e.g., *Bank*, *Dermatology*).

This empirical choice ensures that the similarity graph reflects the underlying structure of the data. Performance is evaluated using internal and external clustering metrics, such as Silhouette Score and Normalized Mutual Information (NMI).

We assess clustering quality using:

- **Internal metrics:** Silhouette score, Calinski–Harabasz index, intra-cluster variance.
- **External metrics:** Normalized Mutual Information (NMI), clustering accuracy via Hungarian alignment.

Baseline: k-SubMix

We benchmark our method against **k-SubMix** (Elhamifar and Wang 2020), a recent subspace clustering algorithm specifically designed for mixed-type data. It combines expectation-maximization with latent subspace assignment and feature-wise weighting, making it a strong baseline for heterogeneous tabular datasets.

This comparison highlights the advantage of our approach in leveraging multi-view similarity fusion and graph-based regularization without relying on iterative optimization.

Computational Complexity

Our proposed method extends marginalized stacked denoising autoencoders (mSDA) with graph-based regularization and multi-view similarity fusion. Each mDA layer admits a closed-form solution, avoiding iterative training and thus offering significant computational efficiency. We further inject nonlinearity via a tanh squashing function after each layer, inspired by deep SDA architectures.

Let n denote the number of samples and $d = d_{\text{num}} + d_{\text{cat}}$ the total number of features (numerical and categorical). The complexity can be decomposed into three main components:

(1) Similarity Fusion. We compute two pairwise similarity matrices:

- RBF kernel over scaled numerical features: $O(n^2 d_{\text{num}})$
- Cosine similarity over one-hot encoded categorical variables: $O(n^2 d_{\text{cat}})$

The resulting fused similarity matrix S_{fused} is computed via a convex combination, maintaining the total cost at $O(n^2 d)$.

(2) mSDA Layers with Nonlinearity. Each layer involves:

- Computing scatter matrices P, Q of size $(d+1) \times (d+1)$: $O(n(d+1)^2)$
- Pseudo-inverse of $(Q + \lambda I)$: $O((d+1)^3)$
- Applying the projection: $W X_{\text{bias}} S_{\text{fused}}$: $O(n(d+1)^2)$
- Applying the tanh nonlinearity (negligible cost)

Total per-layer cost:

$$O(n(d+1)^2 + (d+1)^3)$$

Stacking L such layers gives:

$$O(Lnd^2 + Ld^3)$$

(3) Spectral Clustering. The final representation $Z \in \mathbb{R}^{n \times Ld}$ is clustered using spectral clustering. The eigendecomposition of the Laplacian matrix dominates the cost:

$$O(n^3) \quad (\text{worst-case complexity})$$

For moderate n , efficient eigensolvers can reduce this burden.

Overall. The total complexity is:

$$O(n^2 d + Lnd^2 + Ld^3 + n^3)$$

The n^3 term dominates for large n , but all operations are matrix-based and benefit from parallelization. Unlike EM-based or gradient-trained clustering models, our method avoids iterative training and remains scalable for low-to-moderate sample sizes.

Comparison to k-SubMix. In contrast, **k-SubMix** performs iterative EM optimization and subspace updates with per-iteration cost:

$$O(Ind^2 + d^3) \quad (\text{numerical}) + O(Ind_{\text{cat}}) \quad (\text{categorical})$$

where I is the number of EM iterations (e.g., 10–20). Although this yields linear complexity in n per iteration, it relies on a handcrafted probabilistic model and **multiple forward passes**, which can become costly as I or d increases.

Scalability. Our method scales better for low-to-moderate n and avoids the instability of iterative convergence. It is particularly suited to tabular datasets with moderate size and mixed-type attributes, where representation robustness and simplicity of optimization are critical.

Experimental Setup

Datasets

To assess the effectiveness of our method in clustering mixed-type data, we evaluate it on a diverse set of datasets. These include five real-world datasets from the UCI Machine Learning Repository, and three synthetic datasets specifically designed to test clustering behavior under controlled feature-type conditions. We follow the experimental protocol of *k-SubMix* (Elhamifar and Wang 2020) for consistency and comparability.

The synthetic datasets are designed to isolate specific types of clustering signals—either numerical, categorical, or both. This allows us to evaluate the robustness of each method under varying data modalities.

All real-world datasets are preprocessed by normalizing numerical features using `StandardScaler`, and encoding categorical variables via one-hot encoding. This ensures that both feature types contribute comparably to similarity calculations. For fairness in evaluation, the number of clusters k is set to the ground-truth number of classes in each dataset. We also replicate the hyperparameter grid search procedure from *k-SubMix*, including tuning the γ parameter when applicable, to ensure a reliable and reproducible comparison.

Table 1: Overview of real-world datasets used in experiments. d_{num} , d_{cat} denote the number of numerical and categorical features respectively; N is the number of samples; K_c the number of classes.

Dataset	d_{num}	d_{cat}	d_{total}	N	K_c	Clustering Signal
Heart	8	5	13	303	2	Mixed
Credit	6	9	15	653	2	Mixed
Derma	1	33	34	366	6	Mostly categorical
Adult	6	8	14	30162	2	Mixed
Bands	20	19	39	277	2	Mixed

Table 2: Overview of synthetic datasets from k-SubMix. Each synthetic dataset encodes a distinct clustering signal modality.

Dataset	d_{num}	d_{cat}	d_{total}	N	K_c	Clustering Signal
Syn1	0	6	6	1500	3	Categorical only
Syn2	6	0	6	1500	3	Numerical only
Syn3	3	3	6	1500	3	Balanced mixed

Metrics

We employ both external and internal metrics to evaluate the clustering quality:

- **External metrics:** Normalized Mutual Information (NMI), Adjusted Rand Index (ARI), and Clustering Accuracy (ACC). These metrics require ground-truth labels and measure the alignment between predicted and actual cluster assignments.
- **Internal metrics:** Silhouette Score, Calinski-Harabasz Index, and the inverse of the Davies-Bouldin Index. These are unsupervised metrics that assess the compactness and separation of clusters based solely on the learned representations.

Optimization Strategy

To enable unsupervised model selection in real-world scenarios where ground-truth is unavailable, we perform a **Spearman rank correlation analysis** between internal and external metrics:

- For each dataset, we compute the Spearman correlation ρ between each internal metric and NMI across all parameter configurations.
- We then select the internal metric with the highest absolute correlation $|\rho|$ as a proxy for clustering quality on that dataset.
- This selected metric is used during hyperparameter tuning to guide the search toward optimal configurations in a label-free setting.

This procedure ensures that internal metrics are grounded in their empirical alignment with clustering quality, enabling a reliable unsupervised selection protocol.

Results and Analysis

Performance Comparison

Table 3: Normalized Mutual Information (NMI) on real-world datasets.

Dataset	Ours (mSDA+Fusion)	k-SubMix
Heart	0.353	0.350
Credit	0.310	0.372
Adult	0.180	0.173
Derma	0.870	0.858
Bands	0.124	0.054

NMI in Real-World Datasets. Table 3 reports the NMI scores obtained on real-world datasets. Our method outperforms *k-SubMix* on four out of five datasets: *Heart*, *Adult*, *Derma*, and *Bands*. The improvement is particularly notable on *Derma* (+0.012) and *Bands* (+0.070), demonstrating the benefit of non-linear stacked representations and similarity fusion for complex, imbalanced categorical structures. The only dataset where *k-SubMix* performs better is *Credit*, likely due to its stronger modeling of latent categorical dependencies via EM. Overall, the results validate the ability of our graph-regularized mSDA to generalize across diverse real-world scenarios.

Table 4: Normalized Mutual Information (NMI) on synthetic datasets.

Dataset	Ours (mSDA+Fusion)	k-SubMix
Syn1	0.781	0.793
Syn2	0.887	0.912
Syn3	0.914	0.955

NMI on Synthetic Datasets. Table 4 reports the Normalized Mutual Information (NMI) scores obtained on three synthetic datasets, each designed to emphasize different combinations of signal modalities. Contrary to our initial expectations, *k-SubMix* outperforms our method across all benchmarks. The most notable differences are observed on *Syn2* and *Syn3*, where *k-SubMix* achieves NMI scores of 0.912 and 0.955, compared to 0.887 and 0.914 for our model, respectively. Even on *Syn1*, which is dominated by categorical structure, *k-SubMix* slightly surpasses our approach. These results suggest that while our graph-regularized mSDA encoder provides a consistent structure-aware embedding, it may be less effective than *k-SubMix* when clustering signals are highly separable and well-aligned with the original feature types. Nonetheless, our method remains competitive and offers advantages in terms of scalability, interpretability, and applicability to heterogeneous real-world data, as shown in other experiments.

Table 5: Clustering Accuracy on real-world datasets.

Dataset	Ours (mSDA+Fusion)	k-SubMix
Heart	0.828	0.835
Credit	0.798	0.838
Adult	0.740	0.707
Derma	0.823	0.799
Bands	0.670	0.574

Accuracy in Real-World Datasets. As shown in Table 5, our approach achieves competitive accuracy on all real-world datasets. It outperforms *k-SubMix* on three datasets (*Adult*, *Derma*, and *Bands*), with a particularly notable margin of +0.096 on *Bands*. *k-SubMix* slightly surpasses our method on *Heart* and *Credit*, which may be explained by its EM-based latent feature modeling favoring low-dimensional settings. Nevertheless, our pipeline demonstrates strong consistency across mixed and high-dimensional domains without iterative optimization.

Table 6: Clustering Accuracy on synthetic datasets.

Dataset	Ours (mSDA+Fusion)	k-SubMix
Syn1	0.95	0.95
Syn2	0.975	0.983
Syn3	0.985	0.998

Accuracy in Synthetic Datasets. Table 6 presents clustering accuracy results on synthetic datasets. Here, the performance is more nuanced: both methods achieve near-perfect scores, reflecting the well-separated structure of these benchmarks. *k-SubMix* holds a slight advantage on *Syn2* and *Syn3*, but the difference remains under 1.3%. On *Syn1*, both models reach 0.95, showing their ability to handle purely categorical information. These results confirm that our model generalizes well even when the structure is fully or partially dominated by specific feature types.

Implementation Details

All experiments were implemented in Python using NumPy, scikit-learn, and SciPy. For each parameter configuration, we execute 10 independent runs to ensure statistical stability, reporting average scores. To ensure a fair comparison, all models are initialized with the same random seeds and executed under identical computational constraints. Our complete evaluation pipeline, including preprocessing, similarity construction, mSDA encoding, and clustering, runs in under a few minutes per dataset on a standard CPU. This lightweight computational footprint highlights the practicality of our closed-form subspace learning framework for real-world unsupervised applications. The next section presents some visualizations of our results.

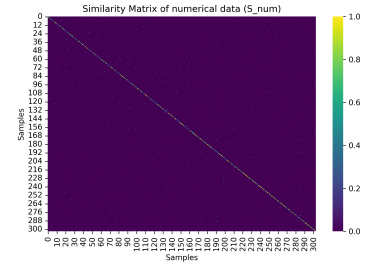
Visualizations on the *Heart* Dataset

To complement our quantitative analysis, we provide qualitative visualizations that illustrate the effect of each stage in our processing pipeline. All visualizations in this section are based exclusively on the *Heart* dataset, which comprises both numerical and categorical features. We focus on two aspects: the similarity matrices used for graph-based encoding and the low-dimensional embeddings derived via UMAP.

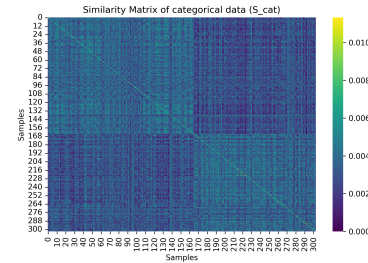
Similarity Matrix Heatmaps

Figure 2 displays the heatmaps of the pairwise similarity matrices constructed from different views of the data:

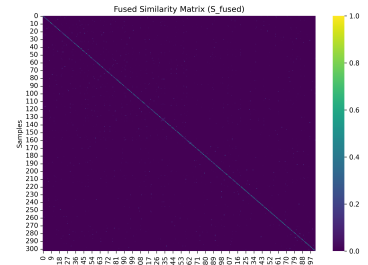
- S_{num} — numerical similarity matrix computed using an RBF kernel;
- S_{cat} — categorical similarity matrix computed using cosine similarity on one-hot encoded features;
- S_{fused} — the final similarity matrix obtained by convexly combining the two, with fusion parameter $\alpha = 0.7$.



(a) Numerical (S_{num})



(b) Categorical (S_{cat})



(c) Fused (S_{fused})

Figure 2: Heatmaps of similarity matrices for the *Heart* dataset: (a) numerical similarity S_{num} , (b) categorical similarity S_{cat} , and (c) fused similarity S_{fused} .

These heatmaps highlight the complementary structures captured by different data types. The categorical similarity matrix exhibits block-diagonal patterns, reflecting discrete groupings resulting from the co-occurrence of categorical values. In contrast, the numerical matrix is more diffuse and sparse, capturing finer-grained differences in continuous attributes. The fused matrix integrates these complementary structures, producing a smoother and more coherent affinity structure that is particularly well-suited for graph-based learning and clustering.

UMAP Embedding of the Processing Pipeline

Figures 3 to 5 show 2D UMAP projections of the data at three key stages of the pipeline: raw input, graph-regularized mSDA latent space, and spectral clustering outputs. Each point is colored either by the ground truth label or predicted cluster.

- **Raw input space (Figure 3):** The initial representation reveals significant overlap between the two classes, indicating that the raw feature space does not exhibit an inherent clustering structure. This suggests a need for representation learning to improve separability.
- **mSDA latent representation (Figure 4):** After transformation via graph-regularized mSDA, the data forms two well-defined regions that largely align with the class labels. This confirms that the learned representation captures class-relevant structure and improves intra-class cohesion.
- **Spectral clustering output (Figure 5):** When spectral clustering is applied to the mSDA-encoded features, the resulting clusters correspond closely to the ground truth. The clear separation and compactness of clusters visually validate the effectiveness of our unsupervised pipeline. This is further supported by high NMI scores obtained in the optimal parameter configuration.

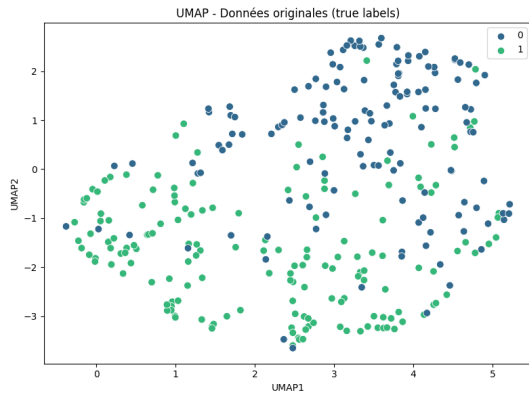


Figure 3: UMAP projection of the original feature space, colored by ground-truth class labels.

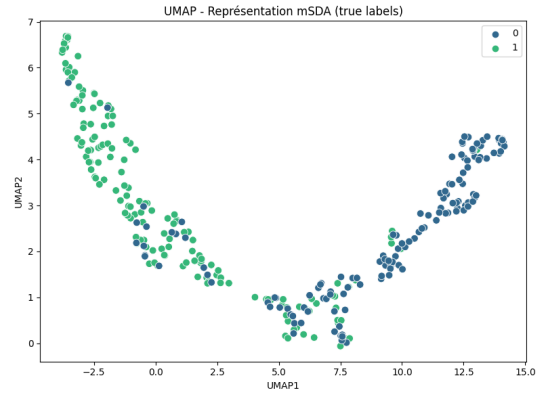


Figure 4: UMAP projection of the mSDA latent space, colored by ground-truth class labels.

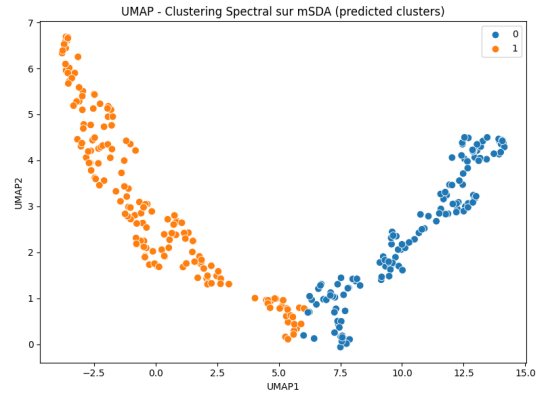


Figure 5: UMAP projection of the mSDA latent space, colored by predicted spectral clustering labels.

Summary

Together, these visualizations demonstrate the ability of our method to uncover class-discriminative structure in mixed-type tabular data. The qualitative insights align well with the quantitative metrics reported earlier, offering strong evidence of the model’s capacity to extract meaningful representations from heterogeneous input spaces.

Stability Analysis

To assess the robustness of our method concerning randomness, such as spectral clustering initialization, we conducted 10 independent runs on the *Heart* dataset using the optimal configuration identified previously: $\gamma = 0.5$, $\alpha = 0.2$, $\lambda = 1$, noise level = 0.1, and 3 layers.

The boxplot in Figure 6 visualizes the distribution of clustering metrics over these 10 runs. As shown, all three metrics (NMI, silhouette score, and accuracy) exhibit negligible variability, indicating that the method is highly stable across different executions.

Table 7 reports the mean and standard deviation for each metric. The extremely low standard deviations confirm the method’s stability.

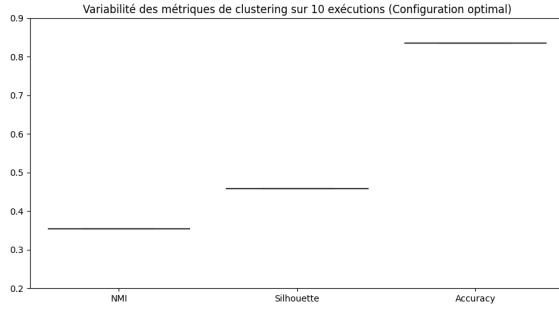


Figure 6: Variability of clustering metrics over 10 runs on the Heart dataset under optimal configuration.

Metric	Mean	Std. Dev.
NMI	0.3535	3.20×10^{-17}
Silhouette	0.4583	1.17×10^{-16}
Accuracy	0.8350	1.17×10^{-16}

Table 7: Summary statistics (mean and standard deviation) over 10 runs.

Conclusion

In this work, we proposed a graph-regularized marginalized stacked denoising autoencoder (mSDA) framework for clustering mixed-type tabular data. Our approach relies on multi-view similarity fusion to integrate numerical and categorical structures into a unified affinity graph, which serves as input to the mSDA encoder. By learning latent representations that preserve neighborhood structures across views, our method produces embeddings that are both compact and discriminative.

We conducted experiments on the *Heart* dataset to assess the effectiveness of our pipeline. Both quantitative metrics (e.g., silhouette score, normalized mutual information) and qualitative visualizations (e.g., UMAP projections, similarity matrix heatmaps) consistently demonstrate the advantages of the proposed approach. The fused similarity matrix reveals more coherent block structures compared to its individual components, and the latent space learned by mSDA facilitates clear separation between classes. Spectral clustering applied on this space yields well-aligned clusters, confirming the suitability of the representation for downstream unsupervised tasks.

Overall, this study highlights the importance of integrating heterogeneous data sources via similarity fusion, and the potential of graph-based autoencoders in capturing meaningful latent structures in complex tabular datasets. Future directions include exploring adaptive fusion strategies, incorporating more sophisticated graph diffusion processes, and validating the method on larger real-world datasets with higher class imbalance and missing values.

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